

Senior Autonomy Integration Engineer - Role-Aligned Resume

Operator-technologist who makes complex systems legible and closes responsibility gaps across hardware, software, operations, quality, and stakeholders. Pittsburgh-based experience spans software-enabled robotics and machine vision, regulated autonomous systems, high-reliability electronics, Python/TensorFlow concepts, root cause, configuration-minded controls, and 24/7 launch execution.

Cross-domain debugging	Root cause and reproduction	Robotics operations	Configuration traceability
Regression and quality systems	Python and TensorFlow	Technical stakeholder alignment	Lean, Gemba, standard work

MOST RELEVANT EXPERIENCE

Vape-Jet

2025-Present

- Director of Operations
- Leads AI-first operating transformation across production, quality, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows in a software-enabled robotics environment.
 - Connects factory, field, customer, and machine-fleet signals from approximately 1,000 fielded robots into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
 - Works directly across recurring machine issues, support evidence, quality concerns, engineering feedback, and operating constraints to make cross-functional problems visible and actionable.

ZeusVu

2018-2022

- Chief Pilot, Autonomous Systems and Aerial Data Operations
- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
 - Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence connecting sensing, mapping, GPS, risk, users, and operating evidence.
 - Designed medical drone delivery and chain-of-custody concepts that translated autonomy, safety, stakeholder authority, and governed execution into an operable system.

Compunetics

2000-2013

- Director of Operations and Engineering Management
- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership in advanced electronics and high-reliability environments.
 - Led engineering, manufacturing, quality, customer delivery, and complex technical programs; built root-cause practices, standard work, performance measures, process controls, and quality systems.
 - Collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD; published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

Amazon

2014-2018

- Operations Management / Site Leader
- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
 - Built labor planning, throughput visibility, escalation paths, standard work, and operating routines across safety, quality, customer experience, and team engagement.
 - Contributed to logistics innovation through Prime Air research and Amazon Flex geolocation concepts.

AI, AUTOMATION AND DISTRIBUTED OPERATIONS

DudeWorth

2017-Present

Founder / AI and Automation Advisor

- Develops AI-readiness and opportunity frameworks that assess value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

Cardinal Building Products

2021-2023

Regional Director

- Built the company's first remote regional operation and modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

Technical toolkit

Python | TensorFlow | SQL | JavaScript
GIS and mapping | Agentic AI | LLM workflows
RAG patterns | Human-in-the-loop
Evaluation and governance | ERP/Odoo | Jira
Confluence | HubSpot | Slack | Tableau

Operating methods

Root cause | Failure reproduction | Quality systems
Process controls | Standard work | Lean
Gemba | Six Sigma | Escalation design
Decision records | Cross-functional alignment
Training and adoption

TECHNICAL LEADERSHIP AND EDUCATION

IEEE - CMPT and EDS Pittsburgh Chapter | Chapter Chair | 2012-2016

Led regional technology programming connecting industry, academia, electronics, manufacturing, STEM outreach, and emerging technology.

University of Pittsburgh | Bachelor of Science

Academic grounding includes materials science, physical chemistry, and biology-related studies. Academic work is indexed at Google Scholar.

Credentials

Google AI Essentials | IBM Enterprise Design Thinking Practitioner | Six Sigma Certification | OSHA 10

SELECTED TECHNICAL CONTRIBUTIONS

Amazon Flex geolocation

Contributed customer-centered geofencing confirmation concepts for delivery execution and proof.

Autonomous aerial inventory

Developed TensorFlow, orthomosaic, GPS, and ground-truth concepts for computer-vision field intelligence.

High-speed electronics

Technical liaison and industry contributor supporting Intel, AMD, and HyperTransport ecosystem work.

Agentic workflow systems

Designs structured-context, retrieval, evaluation, human-authority, escalation, and governance patterns.

LATITUDE AI CONTRIBUTION PROFILE

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|-----------------|---|
| Immediate value | Cross-domain integration triage, Python-enabled tooling, reproducible problem definition, evidence synthesis across hardware and software, configuration traceability, regression thinking, and clear partner-team communication. |
| Operating edge | Moves between system architecture, operating constraints, machine behavior, quality evidence, and ownership boundaries - then converts closure into tests, controls, and reusable mechanisms. |